

Governed Agent Organizations

Persistent identity, context continuity, human authority, and evidence-bounded improvement

Edition: Paper v1.1 - related-work correction to Paper v1

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Status: Paper v1.1 correction candidate; Paper v1 remains frozen at tag `paper-v1` ; publication requires a new reviewed release identity

Abstract

Language models can generate plans, code, analysis, and text, but those abilities alone do not create an organization. A durable organization needs identity that survives model and session changes, working context that is not confused with institutional memory, authority that cannot be inferred from conversation, independent evaluation, explicit ownership and closure, and outcomes that are not reduced to agent activity.

This paper proposes a governed architecture for persistent AI agent organizations. Its contribution is not a single new memory algorithm, context method, orchestration framework, authorization mechanism, runtime policy engine, or optimization technique. It is a system of non-substitutable layers and explicit transitions joining persistent identity, active working context, inspectable institutional memory, human authority, evidence-bounded improvement, organizational coordination, domain engines, and outcome discipline.

The architecture is presented as six falsifiable hypotheses rather than a universal novelty claim. Existing prior art supplies important components. This work asks how those components can be combined without turning context into truth, identity into authority, authorization into behavioral approval, signatures into permission, candidates into active behavior, or activity into outcomes.

A public reference implementation of managed context refresh is available in `wrg32786/aigent-os` at commit `81dd47d3215df0aedb73a378601be431defc7d2f` . One bounded release population observed 11 of 11 transport transactions on one Windows reference seat, with one capsule request, one acknowledgement, one clear, and one fresh session identity per transaction; zero acknowledgements were stranded and zero duplicate clears occurred. This does not establish 20-cycle bookkeeping, fleet reliability, universal platform behavior, implemented organizational products, or business outcomes.

1. Problem and contribution

Capable models do not, by themselves, constitute durable workers or organizations. A production agent system must preserve several different kinds of state and authority across model swaps, context boundaries, process failures, changing evidence, and human decisions.

The central claim of this paper is a governed-integration hypothesis:

```

AgentIdentity != ModelProfile != RuntimeBinding != AuthorityGrant
WorkingContext != InstitutionalMemory
Message != DecisionRecord
Signature != Permission
Candidate != ActivatedCapability
Activity != Outcome

```

The contribution is the refusal to collapse those layers, plus explicit transitions that can fail honestly and be independently reviewed.

What is not claimed as original

This paper does not claim invention of externalized context, memory retrieval, agent workflows, tool protocols, multi-agent collaboration, skill optimization, event-sourced sessions, plugin composition, agent identity management, delegated authorization, runtime policy languages, pre-execution enforcement, cryptographic governance receipts, or standing delegation. Prior work supplies each of those mechanisms or directly adjacent proposals. The proposed contribution is the broader governed combination and its non-substitution rules. Novelty priority for that integration remains unproven.

Evidence vocabulary

Material statements are typed as one of:

```

prior published result
repository-statically-observed mechanism
repository-test-covered mechanism
implemented mechanism
independently demonstrated behavior
repeated operating result
business outcome
inference
hypothesis
UNKNOWN

```

The following substitutions are prohibited:

```

paper result != local result
code present != capability working
test file != executed test
research accepted != implementation
one run != reliability
agent activity != organizational performance
publish event != business outcome

```

2. Six falsifiable hypotheses

H1 - Layer Non-Substitution

Identity, runtime binding, authority, working context, institutional memory, evidence, and outcome are distinct layers. Collapsing any pair is expected to create a predictable failure class.

Examples:

- model selection silently changes privacy or authority;
- a stale checkpoint is treated as current organizational truth;
- a signed message is treated as permission;
- an accepted research result is treated as installed behavior;
- a publish event is treated as a business outcome.

H2 - Continuity Transaction

Continuity is not a checkpoint file or a successful-looking screen. It is a complete transaction:

```
capture requested  
-> request submitted  
-> durable state acknowledged  
-> boundary crossed exactly once  
-> fresh identity observed  
-> prior state re-grounded  
-> wake submitted exactly once  
-> material work continues  
-> transaction closes with evidence
```

Partial transport is not continuity. A pseudo-terminal write is not necessarily a submitted command. A new session is not necessarily a re-grounded session.

A comparative prediction follows from the architecture: discrete capsules plus a fresh-session boundary may reduce cumulative semantic drift relative to repeated summary-of-summary consolidation over many context boundaries. This is a hypothesis, not an empirical superiority claim. DeepSeek Harness retains its complete append-only event log even when its model-visible surface is compacted; that durable log must not be described as lost.

H3 - Governed Improvement

Safe improvement requires separate state and authority for:

candidate generation
 exact artifact identity
 execution environment
 independent evaluation
 promotion decision
 serving scope
 outcome measurement
 rollback or retirement

A skill, harness, engine, evaluator, or organizational change cannot approve itself merely because it improved its own metric.

H4 - Durable Human Gate

Consequential control cannot reliably exist only as chat, an interrupt, a notification, or an approval button. It requires canonical decision objects, authority basis, options, evidence, a durable decision record, propagation, and closure evidence.

A decision is incomplete until dependent work has resumed, changed, delegated, deferred, cancelled, or closed.

H5 - Honest-Terminal Completeness

Systems that cannot terminate honestly as:

UNRUNNABLE
 NO_DATA
 NO_IMPROVEMENT
 HOLD
 REJECT
 ROLLBACK

are structurally biased toward manufacturing progress.

H6 - Research-to-Learning Closure

Research affects an agent organization only through a complete evidence chain:

```
source
-> claim
-> independent disposition
-> retrieval at the relevant task moment
-> bounded use
-> learning candidate
-> pre-registered evaluation
-> approved serving
-> measured outcome
-> retain, revise, retire, or rollback
```

A report stored in memory is not learning. Retrieval is not use. Use is not improvement.

3. Architecture

3.1 Persistent identity

Agent identity belongs outside the current model and session. A model is a runtime binding, not an employee, role, or permission grant. Model changes must preserve identity, mission, memory references, and authority scope, or refuse loudly.

3.2 Active working context

Programmatic context methods may hold source bundles, parsed artifacts, indexes, intermediate results, and child-agent handles. This state is useful but transient and non-authoritative.

3.3 Institutional memory

Institutional memory should remain inspectable, versioned, provenance-bearing, and separable from retrieval accelerators. Summaries and capsules are bounded state carriers, not universal truth. Current operational facts still require current reads from authoritative systems.

A durable capsule trail has different properties from recursive compaction:

- each capsule remains a separate historical artifact;
- later work can identify when a belief first appeared or was superseded;
- current identity does not come from capsule text;
- live re-grounding can reject stale claims;
- retrieval can search the historical trail with provenance and time.

This does not make capsules automatically accurate. A capsule can omit facts, contain stale assumptions, or carry malicious data. It must be framed as historical evidence, not authority.

3.4 Authority

Consequential tools and organizational changes require pre-action authority. Post-action warnings are telemetry, not control. Agent identity, model capability, source authenticity, delegated resource permission, behavioral approval, and action permission remain distinct.

The NIST NCCoE draft concept paper *Accelerating the Adoption of Software and AI Agent Identity and Authorization* (published 2026-02-05) scopes agent identification, authentication, dynamic authorization, delegation, auditing, and non-repudiation as separate enterprise design questions. *AgentSpec: Customizable Runtime Enforcement for Safe and Reliable LLM Agents* (arXiv:2503.18666v3) supplies directly adjacent prior work on external runtime rules evaluated before agent actions. *Behavioral Governance for Autonomous AI Agents: The AgentBound Framework* (arXiv:2606.30970v2) proposes a three-authority behavioral-governance layer, signed policy artifacts, governance receipts, and standing delegation. None of those mechanisms is claimed here as an isolated invention.

3.5 Evidence and evaluation

A green result is evidence only when:

- the population is named;
- exact code, harness, environment, and artifacts are frozen;
- the check can turn red for the intended reason;
- negative tests have same-run positive witnesses;
- the maker is not the sole checker;
- changed identities invalidate prior proof unless the change is outside the proof surface.

3.6 Organizational coordination

Messages are not obligations. Findings are not automatically work. Work requires an authority basis, owner, closure condition, evidence requirement, and reopen path. Organizational truth belongs in canonical objects rather than reconstructed chat.

3.7 Domain engines

Business-function engines should be inspectable packages with explicit objects, events, integrations, source-of-truth maps, authority matrices, privacy rules, budgets, evaluation profiles, and rollback hooks. Generated candidates cannot grant themselves data, credentials, budget, or activation.

3.8 Operator interface

The primary human surface should answer:

What needs me?
 Why?
 What is blocked?
 What is still moving?
 What evidence supports the recommendation?
 What changes if I decide?
 Did dependent work actually resume?

The interface is a view over canonical state. Visual green cannot substitute for freshness, completeness, reconciliation, or authority.

4. Related work and architectural disposition

The source registry records exact pins, licenses or citation-only terms, inspection level, supported claim, and limitations.

Mechanism	Representative source	Disposition	Boundary retained here
Working-context externalization	Recursive Language Models	ADAPT	Working context is not institutional memory or authority
Virtual/tiered context	MemGPT	WATCH/ADAPT	Context management does not create organizational truth
Memory, reflection, planning	Generative Agents; Reflexion; ExpeL; Agent Workflow Memory	ADAPT	Retrieved experience remains evidence, not active authority
Skill optimization	SkillOpt	ADAPT proposal-only	A candidate cannot activate itself
Continual harness refinement	Continual Harness	ADAPT research claims	Optimizer and evaluator remain separate
Agent identity and delegated authorization	<i>Accelerating the Adoption of Software and AI Agent Identity and Authorization</i> (NIST NCCoE draft PDF, 2026-02-05)	ALIGN/WATCH	Identity, authentication, authorization, delegation, and audit are prior concerns; the draft is not a normative standard or demonstrated architecture
Runtime policy enforcement	<i>AgentSpec: Customizable Runtime Enforcement for Safe and Reliable LLM Agents</i> (arXiv:2503.18666v3)	ADAPT	Trigger-predicate-enforcement rules and pre-action controls are prior work; they do not establish persistent organizational identity, memory, or outcome closure
Behavioral governance, receipts, and standing delegation	<i>Behavioral Governance for Autonomous AI Agents: The AgentBound Framework</i> (arXiv:2606.30970v2)	ADAPT/CITE/WATCH	Multi-authority pre-execution governance, signed policy artifacts, verifiable receipts, and persistent delegation are prior proposals; empirical results remain future work
Signed-event collaboration	Buzz	INTEROPERATE	Signature is not authority
Tool/context protocol	MCP	ADAPT	The application retains policy and scope authority
Agent-user event transport	AG-UI	INTEROPERATE	Transport does not grant action authority
Agent-agent protocol	A2A	WATCH	Discovery/task exchange is not decision truth

Mechanism	Representative source	Disposition	Boundary retained here
Minimal agent runtime	Pi agent core; Prime Agent	ADAPT selected seams	Runtime binding does not own identity or authority
Same-session event-sourced runtime and compaction	DeepSeek Harness	INTEROPERATE/ADAPT	Complete log is durable; model-visible recursive summary remains a distinct tradeoff
Dynamic component composition	Cordis and its preprint	ADAPT/CITE/WATCH	Reversible runtime effects are not organizational rollback or authority
Multi-agent software organizations	MetaGPT; ChatDev	WATCH	Role-play and software benchmarks do not establish governed organizations
Kernel-style agent services	AIOS	WATCH/ADAPT	Service composition remains below the governance boundary

Identity, authorization, runtime enforcement, and behavioral governance

The NIST NCCoE draft concept paper *Accelerating the Adoption of Software and AI Agent Identity and Authorization*, published on 2026-02-05, treats agent identification, identity metadata, authentication and key lifecycle, dynamic authorization, least privilege, proof of authority, on-behalf-of delegation, human binding, tamper-verifiable audit, and non-repudiation as distinct enterprise design questions. It is a draft scoping document for potential NCCoE work, not a normative standard, implementation, or demonstrated architecture.

AgentSpec: Customizable Runtime Enforcement for Safe and Reliable LLM Agents ([arXiv:2503.18666v3](#)) establishes directly adjacent prior work for interpretable runtime constraints. Its domain-specific language combines triggers, predicates, and enforcement mechanisms and evaluates constraints during agent execution, including before unsafe actions are dispatched. Its published evaluations are external results; this project has not reproduced them. AgentSpec does not by itself establish persistent organizational identity, delegated authority semantics, institutional memory, decision propagation, or outcome discipline.

Behavioral Governance for Autonomous AI Agents: The AgentBound Framework ([arXiv:2606.30970v2](#)) overlaps more directly with the authority portions of this paper. It proposes delegated authorization, owner-signed and versioned behavioral constitutions, site action contracts, conservative pre-execution composition, cryptographically verifiable governance receipts, and standing delegation for long-running agents. Those mechanisms cannot be claimed here as isolated novelties. The AgentBound paper defines an evaluation framework, but reports comprehensive empirical results and public artifacts as future work; it is therefore treated here as a published architecture proposal rather than demonstrated behavior.

After accounting for this work, the current bounded synthesis claim narrows to a broader governed-integration hypothesis: continuity, persistent identity, working context, institutional memory, authority, organizational obligations, independent improvement promotion, durable human decision closure, and measured outcomes remain non-substitutable and must be connected by explicit transitions. This bounded review does not close the priority question; novelty priority for the complete integration remains unproven.

DeepSeek Harness and Cordis

DeepSeek Harness is a substantial plugin-composed, event-sourced agent runtime. It includes session persistence, goals, plans, tools, permissions, sandboxes, workflows, subagents, skills, and automatic same-session compaction. Cordis provides the underlying service/event/effect composition model.

The architectural lesson is not to rebuild a generic harness blindly. A governed organization can treat DeepSeek Harness, Claude Code, Codex, or another runtime as a `RuntimeBinding`. Identity, authority, institutional memory, accepted evidence, obligations, and outcomes remain outside the replaceable runtime.

Cordis runtime effects can be reversible. Organizational actions often are not. A plugin can unregister a tool; it cannot retroactively undo a payment, revoke a settled decision, or reconcile an external system of record.

5. Context continuity reference implementation

The public `aigent-os` repository ships a managed context-refresh path at commit `81dd47d3215df0aedb73a378601be431defc7d2f`.

The intended sequence is:

```
context pressure rises
-> request one capsule
-> observe one exact acknowledgement
-> submit one clear
-> observe a fresh session identity
-> submit one resume wake
-> load prior work
-> re-ground against live state
-> continue the objective
```

The safety invariant is:

```
no acknowledgement -> no clear
one valid acknowledgement -> exactly one clear
```

The implementation writes slash-command text and Enter separately so the terminal application can use its native prompt queue. Busy or idle output is not a readiness veto after the acknowledgement.

Bounded live result

A reviewed release population on one Windows reference seat produced 11 of 11 observed transport transactions. Each transaction had one capsule request, one acknowledgement, one clear, and one fresh session identity. No acknowledgement was stranded and no duplicate clear occurred.

The accelerated observation protocol later failed its own bookkeeping inside one unchanged session. That failure is not counted as a failed refresh transaction, but it prevents a 20-of-20 claim.

The maximum supported statement is:

One Windows reference seat completed 11 of 11 observed Auto-Refresh transport transactions at the reviewed release snapshot, each with one capsule request, one acknowledgement, one clear, and one fresh session identity, with zero stranded acknowledgements and zero duplicate clears.

The result does not establish fleet reliability, universal platform behavior, or a completed empirical evaluation of material continuation across every cycle. The sanitized case study records the exact public product identity and the evidence boundary.

6. Governance and safety

6.1 Non-self-activation

Candidate generation, evaluation, promotion, installation, serving authority, and outcome measurement are separate transitions. Research acceptance or a green test cannot grant production authority.

6.2 Business-space isolation

Memory, credentials, integrations, budgets, and customer or financial state must not silently cross organizational scopes. Cross-scope access requires explicit authority and a target contract.

6.3 Prompt and memory trust

Persisted text is data, not instruction. A safe renderer should quote, bound, and place untrusted content after policy fences. Semantic influence remains possible even after structural escaping, so provenance and live re-grounding are still required.

6.4 External systems of record

Local memory is not automatically authoritative for customers, accounting, publishing, advertising, contracts, payments, analytics, source code, or live program state.

6.5 Human authority

Humans remain authoritative for consequential activation, publication, finance and capital, credentials, budgets, privacy destinations, cross-scope access, organizational policy, and acceptance of material risk.

7. Evidence maturity

Demonstrated

- a bounded managed Auto-Refresh transport population on one Windows reference seat;
- public product code and direct test coverage for managed refresh;

- a one-command supported install path for the managed runner.

Repository-observed or test-covered

- deterministic capsule selection and resume rendering;
- current-session identity supplied by the fresh session-start event rather than historical capsule text;
- protected two-part terminal command submission;
- explicit non-claims and failure terminals;
- reusable program-control artifacts in this repository.

Proposed or held

- a new general kernel substrate;
- a full operator Studio;
- generated Domain Engines;
- a runnable reference corporation;
- automated evidence-backed self-improvement;
- broad fleet conformance.

Unknown or unestablished

- 20 consecutive independently verified cycles;
- fleet reliability;
- universal platform reliability;
- lower operator burden;
- improved revenue, retention, customer, or financial outcomes;
- empirical superiority of fresh-session capsules over recursive compaction.

8. Evaluation program

A credible evaluation separates:

1. continuity and refresh;
2. memory retrieval and temporal truth;
3. prompt and authority boundaries;
4. skill and harness improvement;
5. evaluator integrity;
6. model/runtime swaps;
7. agent coordination and obligations;
8. operator comprehension and burden;
9. privacy and business-space isolation;
10. customer, financial, and business outcomes.

No opaque composite score may average away authority, privacy, stale-state, rollback, or business-outcome failure.

A future comparison of recursive compaction and fresh-session capsules should hold the task and model population constant, inject exact facts, later corrections, uncertainty, and stale decisions, then score retention, provenance, stale-fact rejection, next-action accuracy, and token cost over many boundaries.

9. Limitations and non-claims

This paper does not establish:

- novelty priority for every hypothesis;
- novelty of agent identity, delegated authorization, runtime policy enforcement, governance receipts, or standing delegation;
- 20-cycle or fleet reliability;
- universal prompt-injection resistance;
- safe autonomous self-improvement;
- implemented multi-business isolation;
- an implemented operator cockpit;
- implemented Domain Engines;
- a runnable reference corporation;
- lower operator burden;
- improved business outcomes.

The related-work population remains bounded and is not a systematic literature review. The sources added in Paper v1.1 caused the current bounded synthesis claim to narrow; additional identity, authorization, runtime-governance, behavioral-contract, provenance, and auditability work may narrow it further.

The detailed raw operating evidence for the bounded live result is not published in this edition. The public claim is limited to the sanitized extract and publicly inspectable product identity. The absence of raw private evidence is a limitation, not permission to infer a broader result.

The Cordis preprint has no reuse license identified at the pinned repository state. It is cited and paraphrased only; no prose, figures, diagrams, pages, or code are reproduced.

10. Conclusion

Capable agents do not become organizations by adding more context, memory, tools, agents, or autonomy. They become organizations only when persistent identity, inspectable institutional memory, authority, independent evaluation, obligations, rollback, and outcomes are joined by explicit, falsifiable transitions.

This paper does not claim invention of agent identity, delegated authorization, runtime policy enforcement, behavioral-governance receipts, or standing delegation. The current bounded synthesis claim is the broader governed integration and a set of failure-derived hypotheses. Novelty priority remains unproven. A bounded context-refresh transport result exists, while repeated reliability, broad implementation, organizational performance, and business outcomes remain open empirical questions.